In Exercises 1–2, estimate the minimum number of subintervals needed to approximate the integrals with an error of magnitude less than10^{-4} by (a) the Trapezoidal Rule and (b) Simpson's Rule.

1. (0.1 point) $\int_0^2 \sin(x+1)\,dx$
2. (0.1 point) $\int_{-1}^1 \cos(x+\pi)\,dx$

Which of the sequences $\{a_n\}$ in Exercises 3–5 converge, and which diverge? Find the limit of each convergent sequence.

3. (0.1 point) $a_n = \frac{(-4)^n}{n!}$
4. (0.1 point) $a_n = \left(\frac{3n+1}{3n-1}\right)^n$
5. (0.1 point) $a_n = \frac{(10/11)^n}{(9/10)^n+(11/12)^n}$

6. (0.1 point) Show by example that $\sum a_n/b_n$ may diverge even though $\sum a_n$ and $\sum b_n$ converge and no b_n equals 0.
7. (0.1 point) Find convergent geometric series $A = \sum a_n$ and $B = \sum b_n$ that illustrate the fact that $\sum a_n b_n$ may converge without being equal to AB.
8. (0.1 point) Show by example that $\sum a_n/b_n$ may converge to something other than A/B even when $A = \sum a_n, B = \sum b_n, n \neq 0$, and no b_n equals 0.

In Exercises 9–11, use any method to determine if the series con verges or diverges. Give reasons for your answer.

9. (0.1 point) $\sum_{n=1}^{\infty} \frac{(-3)^n}{n^3 \cdot 2^n}$
10. (0.1 point) $\sum_{n=1}^{\infty} \frac{(n!)^2}{(2n)!}$
11. (0.1 point) $\sum_{n=1}^{\infty} \frac{(2n+3)(2^n+3)}{3^n+2}$

In Exercises 12-14, (a) find the series' radius and interval of convergence. For what values of x does the series converge (b) absolutely, (c) conditionally?

12. (0.1 point) $\sum_{n=1}^{\infty} \frac{1}{2\cdot 4\cdot 6\ldots(2n)} \cdot x^n$
13. (0.1 point) $\sum_{n=1}^{\infty} \frac{3\cdot 5\cdot 7\ldots(2n+1)}{n^2\cdot 2^n} x^{n+1}$
14. (0.1 point) $\sum_{n=1}^{\infty} \left(\sqrt{n+1} - \sqrt{n}\right)(x-3)^n$

15. (0.1 point) Use Taylor's formula with a=0 and n=3 to find the standard cubic approximation of $f(x) = \frac{1}{1-x}$ $at\ x = 0$. Give an upper bound for the magnitude of the error in the approximation when $|x| \leq 0.1$.
16. (0.1 point) Show that if the graph of a twice-differentiable function ƒ(x) has an inflection point at x=a, then the linearization of ƒ at x=a is also the quadratic approximation of ƒ at x=a.

In Exercises 17-18, find the Taylor series generated by ƒ at x=a

17. (0.1 point) $f(x) = \cos\left(2x + \frac{\pi}{2}\right), a = \frac{\pi}{4}$
18. (0.1 point) $f(x) = \sqrt{x+1}, a = 0$

Exercises 19–21 give parametric equations and parameter intervals for the motion of a particle in the xy-plane. Identify the particle's path by finding a Cartesian equation for it**. Graph the Cartesian equation.** Indicate the portion of the graph traced by the particle and the direction of motion.

19. (0.1 point) $x = 3t; y = 9t^2;\ -\infty < t < +\infty$

20. (0.1 point) $x = -\sqrt{t}; y = t; t \geq 0$

21. (0.1 point) $x = 2t - 5; y = 4t - 7;\ -\infty < t < +\infty$

In Exercises 22–24, r(t) is the position of a particle in space at time t. Find the particle's velocity and acceleration vectors. Then find the particle's speed and direction of motion at the given value of t. Write the particle's velocity at that time as the product of its speed and direction.

22. (0.1 point) $r(t) = (t+1)i + (t^2 - 1)j + 2tk; t = 1$

23. (0.1 point) $r(t) = (1+t)i + \frac{t^2}{\sqrt{2}}j + \frac{t^3}{3}k; t = 1$

24. (0.1 point) $r(t) = (2\cos t)i + (3\sin t)j + 4tk; t = \frac{\pi}{2}$

Find the limits in Exercises 25–28 by rewriting the fractions first.

25. (0.1 point) $\lim\limits_{(x,y)\to(0,0)} \frac{\sin(x^2+y^2)}{x^2+y^2}$

26. (0.1 point) $\lim\limits_{(x,y)\to(0,0)} \frac{1-\cos(xy)}{xy}$

27. (0.1 point) $\lim\limits_{(x,y)\to(1,-1)} \frac{x^3+y^3}{x+y}$

28. (0.1 point) $\lim\limits_{(x,y)\to(2,2)} \frac{x-y}{x^4-y^4}$

Find all the second-order partial derivatives of the functions in Exercises 29–31.

29. (0.1 point) $h(x,y) = xe^y + y + 1$

30. (0.1 point) $r(x,y) = \ln(x+y)$

31. (0.1 point) $s(x,y) = \tan^{-1}\left(\frac{y}{x}\right)$

In Exercises 32-33, sketch the curve $f(x,y) = c$ together with ∇f and the tangent line at the given point. Then write an equation for the tangent line.

32. (0.1 point) $xy = -4,\ (2;-2)$

33. (0.1 point) $x^2 - xy + y^2 = 7,\ (-1;2)$

In Exercises 34–35, find ∇f at the given point.

34. (0.1 point) $f(x,y,z) = (x^2+y^2+z^2)^{-\frac{1}{2}} + \ln(xyz)\,; (-1;2;-2)$

35. (0.1 point) $f(x,y,z) = e^{x+y}\cdot\cos z\ + (y+1)\cdot\sin^{-1}x\,; \left(0;0;\frac{\pi}{6}\right)$

In Exercises 36–38, find the absolute maxima and minima of the functions on the given domains.

36. (0.1 point) $f(x,y) = 2x^2 - 4x + y^2 - 4y + 1$ on the closed triangular plate bounded by the lines $x = 0, y = 2, y = 2x$ in the first quadrant

37. (0.1 point) $D(x,y) = x^2 - xy + y^2 + 1$ on the closed triangular plate in the first quadrant bounded by the lines $x = 0, y = 4, y = x$

38. (0.1 point) $f(x,y) = x^2 + y^2$ on the closed triangular plate bounded by the lines $x = 0, y = 0, y + 2x = 2$ in the first quadrant

39. (0.1 point) Minimum distance to the origin. Find the point on the surface $z = xy + 1$ nearest the origin.

40. (0.1 point) Minimum distance to the origin. Find the points on the surface $z^2 = xy + 4$ closest to the origin.

41. (0.1 point) Minimum distance to the origin. Find the point(s) on the surface $xyz = 1$ closest to the origin.

42. (0.1 point) Find the volume of the region bounded above by the elliptical paraboloid $z = 16 - x^2 - y^2$ and below by the square $R: 0 \le x \le 2, 0 \le y \le 2$.

43. (0.1 point) Find the volume of the region bounded above by the plane $z = 2 - x - y$ and below by the square $R: 0 \le x \le 1,\ 0 \le y \le 1$.

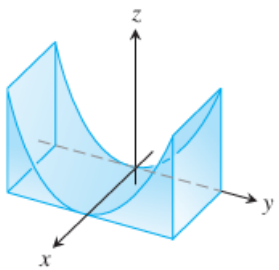
In Exercises 44–45, integrate f over the given region.

44. (0.1 point) Quadrilateral $f(x, y) = x/y$ over the region in the first quadrant bounded by the lines $y = x, y = 2x, x = 1, and\ x = 2$

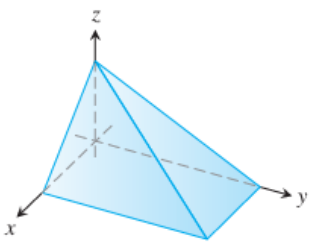
45. (0.1 point) Triangle $f(x, y) = x^2 + y^2$ over the triangular region with vertices $(0, 0), (1, 0), and\ (0, 1)$

Find the volumes of the regions in Exercises 46–50.

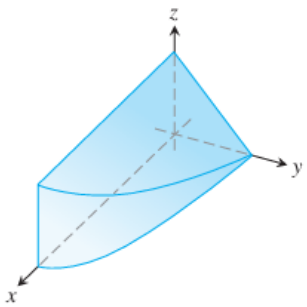
46. (0.1 point) The region between the cylinder $z = y^2$ and the xy-plane that is bounded by the planes $x = 0, x = 1, y = -1, y = 1$



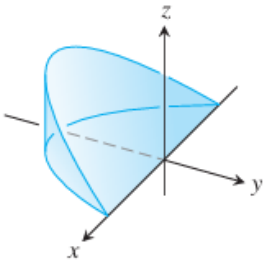
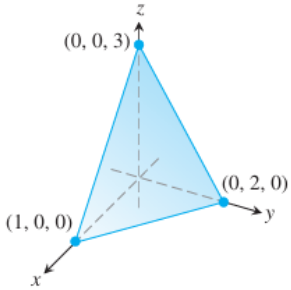
47. (0.1 point) The region in the first octant bounded by the coordinate planes and the planes $x + z = 1,\ \ y + 2z = 2$



48. (0.1 point) The region in the first octant bounded by the coordinate planes, the plane $y + z = 2$, and the cylinder $x = 4 - y^2$



49. (0.1 point) The wedge cut from the cylinder $x^2 + y^2 = 1$ by the planes $z = -y$ and $z = 0$



50. (0.1 point) The tetrahedron in the first octant bounded by the coordinate planes and the plane passing through (1, 0, 0), (0, 2, 0), and (0, 0, 3)